

Homework 3

35 points

1. A farm has chickens, cows, and pigs. Total animals = 50. Total legs = 140. Also there are twice as many chickens as cows. How many of each? **10 points**

Solve using Gauss Jordan Elimination.

2. Is the following system inconsistent, dependent or will it lead to unique solutions?

$$x + 2y - z = 1$$

$$2x + 4y - 2z = 2$$

$$3x + 6y - 3z = 5$$

Show your work using Gauss-Jordan Elimination. **10 points**

3. Let A be a $3 \times k$ matrix and B be a 2×4 matrix. For what value of k will the matrix multiplication be defined and what will the dimensions/order of the resultant matrix be? (*Show your reasoning*) **2 points**

4. Calculate the angle between the following vectors: $\vec{a} = (1,2)$ and $\vec{b} = (3,4)$
3 points

5. Perform matrix multiplication between the following matrices, if valid. If not, explain why not: **5 * 2 = 10 points**

1.

$$\begin{bmatrix} 5 & -4 & -2 \\ 5 & -5 & 4 \\ 2 & 5 & -4 \\ -5 & 4 & 3 \\ 3 & -4 & -3 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$$

2.

$$\begin{bmatrix} 0 & -1 & -1 & 3 \\ 5 & -5 & -2 & 2 \\ 1 & 0 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 & -3 \\ -2 & -1 \\ 3 & -3 \end{bmatrix}$$